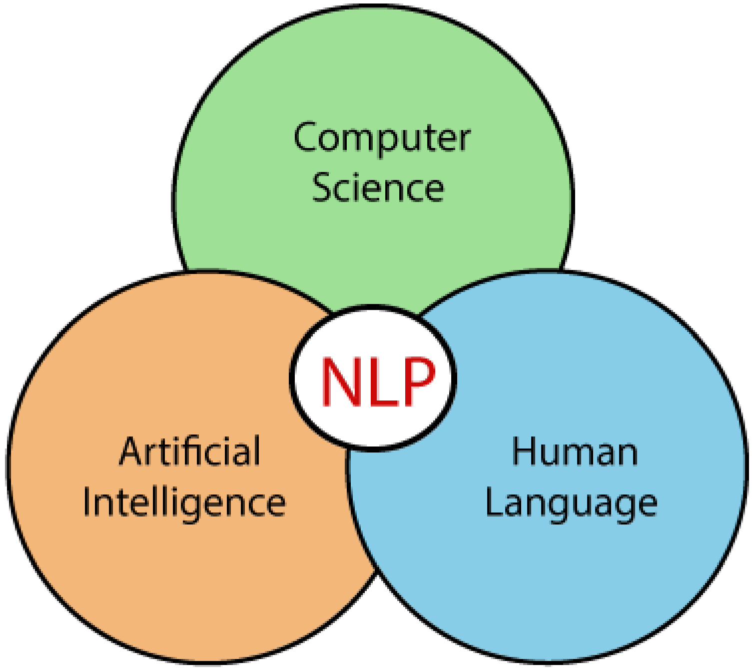
Unit-1 Introduction to NLP

What is Natural Language Processing (NLP)

- Natural Language Processing (NLP) is a subfield of Artificial Intelligence (AI) that combines the power of linguistics and computer science to study the rules and structure of language, and create intelligent systems capable of understanding, analyzing, and extracting meaning from text and speech.



- The process of computer analysis of input provided in a human language (natural language), and conversion of this input into a useful form of representation.

- In Natural Language Processing (NLP), the goal is to make computers understand the unstructured text and retrieve meaningful pieces of information from it.
- The field of NLP is primarily concerned with getting computers to perform useful and interesting tasks with human languages.
- The field of NLP is secondarily concerned with helping us come to a better understanding of human language.

Example:

Take **Sentiment Analysis**, for example, which uses natural language processing to detect emotions in text. This classification task is one of the most popular tasks of NLP, often used by businesses to automatically detect brand sentiment on social media.

Analyzing these interactions can help brands detect urgent customer issues that they need to respond to right away, or monitor overall customer satisfaction.

Why natural language processing is important?

- The amount of data generated by us keep increasing by the day, raising the need for analyzing and documenting this data.
- NLP enables computers to read this data and convey the same in languages humans understand.
- From medical records to government data, a lot of these data is unstructured. NLP helps computers to put them in proper formats. Once that is done, computers analyze texts and speech to extract meaning. Not only is the process automated, but also near-accurate all the time.

What are the Benefits of NLP

There are many benefits of NLP, but here are just a few top-level benefits:

- **Perform large-scale analysis:** Natural Language Processing helps machines automatically understand and analyze huge amounts of unstructured text data, like social media comments, customer support tickets, online reviews, news reports, and more.
- **Automate processes in real-time:** Natural language processing tools can help machines learn to sort and route information with little to no human interaction – quickly, efficiently, accurately, and around the clock.
- **Tailor NLP tools to your industry:** Natural language processing algorithms can be tailored to your needs and criteria, like complex, industry-specific language – even sarcasm and misused words.

Brief History of NLP

- 1940s –1950s: Foundations
  - Development of formal language theory (Chomsky, Backus, Naur, Kleene)
  - Probabilities and information theory (Shannon)
- 1957 – 1970s:
  - Use of formal grammars as basis for natural language processing (Chomsky, Kaplan)
  - Use of logic and logic based programming (Minsky, Winograd, Colmerauer, Kay)
- 1970s – 1983:
  - Probabilistic methods for early speech recognition (Jelinek, Mercer)
  - Discourse modeling (Grosz, Sidner, Hobbs)
- 1983 – 1993:
  - Finite state models (morphology) (Kaplan, Kay)
- 1993 – present:
  - Strong integration of different techniques, different areas.

Challenges of Natural Language Processing

NLP is a powerful tool with huge benefits, but there are still a number of Natural Language Processing limitations and problems:

1. Contextual words and phrases and homonyms
2. Synonyms
3. Irony and sarcasm
4. Ambiguity
5. Errors in text or speech
6. Colloquialisms and slang
7. Domain-specific language
8. Low-resource languages
9. Lack of research and development

1. Contextual words and phrases and homonyms

- The same words and phrases can have different meanings according the context of a sentence and many words – especially in English – have the exact same pronunciation but totally different meanings.
- For example: I **ran** to the store because we **ran** out of milk.

- These are easy for humans to understand because we read the context of the sentence and we understand all of the different definitions. And, while NLP language models may have learned all of the definitions, differentiating between them in context can present problems.
- **Homonym:** two or more words that are pronounced the same but have different definitions – can be problematic for question answering and speech-to-text applications because they aren't written in text form. Usage of their and there, for example, is even a common problem for humans.

2. Synonyms

- Synonyms can lead to issues similar to contextual understanding because we use many different words to express the same idea.

- Furthermore, some of these words may convey exactly the same meaning, while some may be levels of complexity (small, little, tiny) and different people use synonyms to denote slightly different meanings within their personal vocabulary.

- So, for building NLP systems, it’s important to include all of a word’s possible meanings and all possible synonyms.

- Text analysis models may still occasionally make mistakes, but the more relevant training data they receive, the better they will be able to understand synonyms.

3. Irony and sarcasm

- Irony and sarcasm present problems for machine learning models because they generally use words and phrases that, strictly by definition, may be positive or negative.
- For example: how do you teach a machine to understand an expression that’s used to say the opposite of what’s true?

4. Ambiguity

- Ambiguity in NLP refers to sentences and phrases that potentially have two or more possible interpretations.

    - **Lexical ambiguity:** a word that could be used as a verb, noun, or adjective.
    - **Semantic ambiguity:** the interpretation of a sentence in context. For example: I saw the boy on the beach with my binoculars. This could mean that I saw a boy through my binoculars or the boy had my binoculars with him
    - **Syntactic ambiguity:** In the sentence above, this is what creates the confusion of meaning. The phrase with my binoculars could modify the verb, "saw," or the noun, "boy."

Even for humans this sentence alone is difficult to interpret without the context of surrounding text. POS (part of speech) tagging is one NLP solution that can help solve the problem, somewhat.

5. Errors in text and speech

- Misspelled or misused words can create problems for text analysis.

- Autocorrect and grammar correction applications can handle common mistakes, but don’t always understand the writer’s intention.

- With spoken language, mispronunciations, different accents, stutters, etc., can be difficult for a machine to understand.

- However, as language databases grow and smart assistants are trained by their individual users, these issues can be minimized.

6. Colloquialisms and slang

- Informal phrases, expressions, idioms, and culture-specific lingo present a number of problems for NLP – especially for models intended for broad use.

- Because as formal language, colloquialisms may have no "dictionary definition" at all, and these expressions may even have different meanings in different geographic areas. Furthermore, cultural slang is constantly morphing and expanding, so new words pop up every day.

- This is where training and regularly updating custom models can be helpful, although it oftentimes requires quite a lot of data.

7. Domain-specific language

- Different businesses and industries often use very different language.
- An NLP processing model needed for healthcare, for example, would be very different than one used to process legal documents.
- These days, however, there are a number of analysis tools trained for specific fields, but extremely niche industries may need to build or train their own models.

8. Low-resource languages

- AI machine learning NLP applications have been largely built for the most common, widely used languages.
- However, many languages, especially those spoken by people with less access to technology often go overlooked and under processed.
- For example, by some estimations, there are over 3,000 languages in Africa, alone. There simply isn’t very much data on many of these languages.

9. Lack of research and development

- Machine learning requires A LOT of data to function to its outer limits – billions of pieces of training data.
- The more data NLP models are trained on, the smarter they become. That said, data is only growing by the day, as are new machine learning techniques and custom algorithms. All of the problems above will require more research and new techniques in order to improve on them.

Current challenges in NLP

1. Breaking sentences into tokens.
2. Tagging parts of speech (POS).
3. Building an appropriate vocabulary.
4. Linking the components of a created vocabulary.
5. Understanding the context.
6. Extracting semantic meaning.
7. Named Entity Recognition (NER).
8. Transforming unstructured data into structured data.
9. Ambiguity in speech.

Forms of Natural Language

- The input/output of a NLP system can be:
  - **written text**
  - **speech**
- We will mostly concerned with written text (not speech).
- To process written text, we need:
  - **lexical, syntactic, semantic knowledge about the language**
  - **discourse information, real world knowledge**
- To process spoken language, we need everything required to process written text, plus the challenges of speech recognition and speech synthesis.

Components of NLP

- **Natural Language Understanding:** Natural Language Understanding (NLU) helps the machine to understand and analyze human language by extracting the metadata from content such as concepts, entities, keywords, emotion, relations, and semantic roles.

- NLU involves the following tasks -
  - Mapping the given input in the natural language into a useful representation.
  - Different level of analysis required:

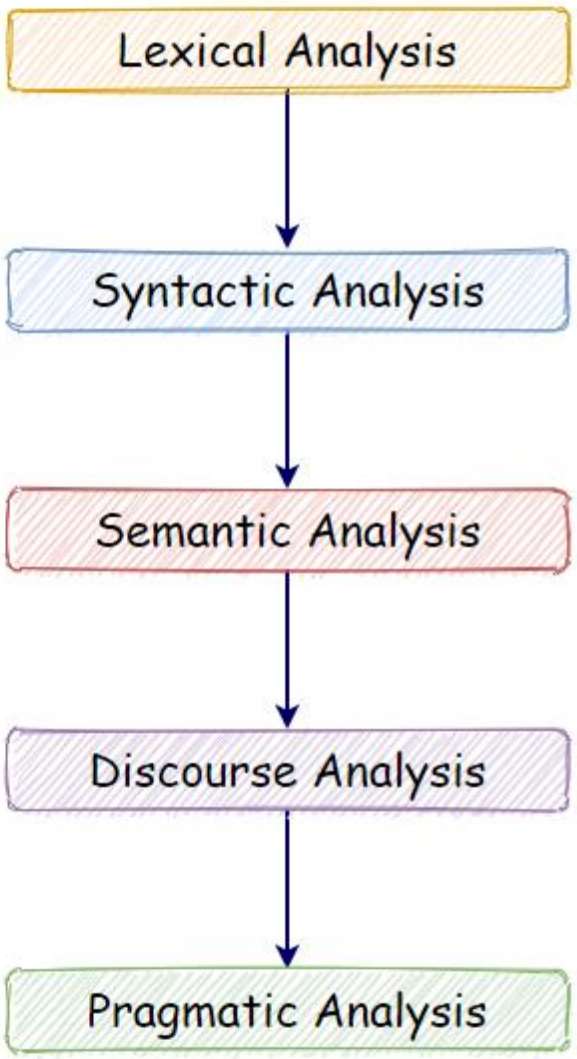
    morphological analysis,

    syntactic analysis,

    semantic analysis,

    discourse analysis,

    pragmatic analysis


Lexical Analysis
Syntactic Analysis
Semantic Analysis
Discourse Analysis
Pragmatic Analysis

- **Natural Language Generation:**

  Natural Language Generation (NLG) acts as a translator that converts the computerized data into natural language representation. It mainly involves Text planning, Sentence planning, and Text Realization.

  – Producing output in the natural language from some internal representation.

  – Different level of synthesis required:

    deep planning (what to say),

    syntactic generation

NLP vs NLU vs. NLG

- **Natural language processing (NLP)** seeks to convert unstructured language data into a structured data format to enable machines to understand speech and text and formulate relevant, contextual responses. Its subtopics include natural language processing and natural language generation.
- **Natural language understanding (NLU)** focuses on machine reading comprehension through grammar and context, enabling it to determine the intended meaning of a sentence.
- **Natural language generation (NLG)** focuses on text generation, or the construction of text in English or other languages, by a machine and based on a given dataset.

NLU	NLP	NLG
NLU is a narrow concept.	NLP is a wider concept.	NLU is a narrow concept.
If we only talk about an understanding text, then NLU is enough.	But if we want more than understanding, such as decision making, then NLP comes into play.	It generates a human-like manner text based on the structured data.
NLU is a subset of NLP.	NLP is a combination of NLU and NLG for conversational Artificial Intelligence problems.	NLU is a subset of NLP.
It is not necessarily that what is written or said is meant to be the same. There can be flaws and mistakes. NLU makes sure that it will infer correct intent and meaning even data is spoken and written with some errors. It is the ability to understand the text.	But, if we talk about NLP, it is about how the machine processes the given data. Such as make decisions, take actions, and respond to the system. It contains the whole End to End process. Every time NLP doesn't need to contain NLU.	NLU generates structured data, but it is not necessarily that the generated text is easy to understand for humans. Thus NLG makes sure that it will be human-understandable.
It reads data and converts it to structured data.	NLP converts unstructured data to structured data.	NLG writes structured data.

Knowledge of Language

- **Phonology** – concerns how words are related to the sounds that realize them.
- **Morphology** – concerns how words are constructed from more basic meaning units called morphemes. A morpheme is the primitive unit of meaning in a language.
- **Syntax** – concerns how can be put together to form correct sentences and determines what structural role each word plays in the sentence and what phrases are subparts of other phrases.
- **Semantics** – concerns what words mean and how these meaning combine in sentences to form sentence meaning. The study of context-independent meaning.

- **Pragmatics** – concerns how sentences are used in different situations and how use affects the interpretation of the sentence.

- **Discourse** – concerns how the immediately preceding sentences affect the interpretation of the next sentence. For example, interpreting pronouns and interpreting the temporal aspects of the information.

- **World Knowledge** – includes general knowledge about the world. What each language user must know about the other’s beliefs and goals.

Why NL Understanding is hard?

- Natural language is extremely rich in form and structure, and **very ambiguous**.
  - How to represent meaning,
  - Which structures map to which meaning structures.
- One input can mean many different things. Ambiguity can be at different levels.
  - Lexical (word level) ambiguity -- different meanings of words
  - Syntactic ambiguity -- different ways to parse the sentence
  - Interpreting partial information -- how to interpret pronouns
  - Contextual information -- context of the sentence may affect the meaning of that sentence.
- Many input can mean the same thing.
- Interaction among components of the input is not clear.

Ambiguity and Uncertainty in Language

I made her duck.

- How many different interpretations does this sentence have?
- What are the reasons for the ambiguity?
- The categories of knowledge of language can be thought of as ambiguity resolving components.
- How can each ambiguous piece be resolved?
- Does speech input make the sentence even more ambiguous?
  - Yes – deciding word boundaries

- Some interpretations of : **`I made her duck.`**
  1. I cooked *duck* for her.
  2. I cooked *duck* belonging to her.
  3. I created a toy duck which she owns.
  4. I caused her to quickly lower her head or body.
  5. I used magic and turned her into a *duck*.

- `duck` – morphologically and syntactically ambiguous: noun or verb.
- `her` – syntactically ambiguous: dative or possessive.
- `make` – semantically ambiguous: cook or create.
- `make` – syntactically ambiguous:

Types of ambiguities

1. **Lexical Ambiguity:** The ambiguity of a single word is called lexical ambiguity. For example, treating the word silver as a noun, an adjective, or a verb.

2. **Syntactic Ambiguity:** This kind of ambiguity occurs when a sentence is parsed in different ways.

   For example, the sentence "The man saw the girl with the telescope". It is ambiguous whether the man saw the girl carrying a telescope or he saw her through his telescope.

3. **Semantic Ambiguity:** This kind of ambiguity occurs when the meaning of the words themselves can be misinterpreted. In other words, semantic ambiguity happens when a sentence contains an ambiguous word or phrase.

   For example, the sentence "The car hit the pole while it was moving" is having semantic ambiguity because the interpretations can be "The car, while moving, hit the pole" and "The car hit the pole while the pole was moving".

4. **Anaphoric Ambiguity:** This kind of ambiguity arises due to the use of anaphora entities in discourse.

   For example, the horse ran up the hill. It was very steep. It soon got tired. Here, the anaphoric reference of "it" in two situations cause ambiguity.

5. **Pragmatic ambiguity:** Such kind of ambiguity refers to the situation where the context of a phrase gives it multiple interpretations. In simple words, we can say that pragmatic ambiguity arises when the statement is not specific.

   For example, the sentence "I like you too" can have multiple interpretations like I like you (just like you like me), I like you (just like someone else dose).

Resolve Ambiguities

- We will introduce *models* and *algorithms* to resolve ambiguities at different levels.
- **part-of-speech tagging --** Deciding whether `duck` is verb or noun.
- **word-sense disambiguation --** Deciding whether `make` is `create` or `cook`.
- **lexical disambiguation --** Resolution of part-of-speech and word-sense ambiguities are two important kinds of lexical disambiguation.
- **syntactic ambiguity --** `her duck` is an example of syntactic ambiguity, and can be addressed by probabilistic parsing.

Models to Represent Linguistic Knowledge

- We will use certain formalisms (*models*) to represent the required linguistic knowledge.
- **State Machines** – FSAs(Finite State Automata), FSTs(Finite State Transducers), HMMs, ATNs (Augmented Transition networks), RTNs(Recursive Transition networks)
- **Formal Rule Systems** -- Context Free Grammars, Unification Grammars, Probabilistic CFGs.
- **Logic-based Formalisms** -- First order predicate logic, some higher order logic.
- **Models of Uncertainty** -- Bayesian probability theory.

Algorithms to Manipulate Linguistic Knowledge

- We will use *algorithms* to manipulate the models of linguistic knowledge to produce the desired behavior.
- Since the language is ambiguous at all levels,

  these algorithms are never simple processes.
- Categories of most algorithms that will be used can fall into following categories.
  - state space search
  - dynamic programming

Language and Intelligence

<u>Turing Test</u>

Computer *Human*

Human Judge

- *Human Judge* asks tele-typed questions to *Computer* and *Human*.
- *Computer's* job is to act like a human.
- *Human's* job is to convince Judge that he is not machine.
- *Computer* is judged "intelligent" if it can fool the judge
- Judgment of intelligence is linked to appropriate answers to questions from the system.

NLP - an inter-disciplinary Field

- NLP borrows techniques and insights from several disciplines.
- **Linguistics**: How do words form phrases and sentences? What constraints the possible meaning for a sentence?
- **Computational Linguistics**: How is the structure of sentences are identified? How can knowledge and reasoning be modeled?
- **Computer Science**: Algorithms for automations, parsers.
- **Engineering**: Stochastic techniques for ambiguity resolution.
- **Psychology**: What linguistic constructions are easy or difficult for people to learn to use?
- **Philosophy**: What is the meaning, and how do words and sentences acquire it?

Some Buzz-Words

- NLP – Natural Language Processing
- CL – Computational Linguistics
- SP – Speech Processing
- HLT – Human Language Technology
- NLE – Natural Language Engineering
- SNLP – Statistical Natural Language Processing
- Other Areas:
  - Speech Generation, Text Generation, Speech Understanding, Information Retrieval,
  - Dialogue Processing, Inference, Spelling Correction, Grammar Correction,
  - Text Summarization, Text Categorization,

Natural Language Understanding

↓ Words

Morphological Analysis

↓ Morphologically analyzed words *(another step: POS tagging)*

Syntactic Analysis

↓ Syntactic Structure

Semantic Analysis

↓ Context-independent meaning representation

Discourse Processing

↓ Final meaning representation

Natural Language Generation

Meaning representation

Utterance Planning

Meaning representations for sentences

Sentence Planning and Lexical Choice

Syntactic structures of sentences with lexical choices

Sentence Generation

Morphologically analyzed words

Morphological Generation

Words

Morphological Analysis

- Analyzing words into their linguistic components (morphemes).
- Morphemes are the smallest meaningful units of language.

cars	car+PLU
giving	give+PROG

- Ambiguity: More than one alternatives

flies	fly_{VERB}+PROG
	fly_{NOUN}+PLU

Part-of-Speech (POS) Tagging

- Each word has a part-of-speech tag to describe its category.
- Part-of-speech tag of a word is one of major word groups (or its subgroups).
  - **open classes --** noun, verb, adjective, adverb
  - **closed classes --** prepositions, determiners, conjuctions, pronouns, particples
- POS Taggers try to find POS tags for the words.
- `duck` is a verb or noun? (morphological analyzer cannot make decision).
- A POS tagger may make that decision by looking the surrounding words.
  - `Duck!` (verb)
  - `Duck is delicious for dinner.` (noun)

Lexical Processing

- The purpose of lexical processing is to determine meanings of individual words.
- Basic methods is to lookup in a database of meanings -- **lexicon**
- We should also identify non-words such as punctuation marks.
- Word-level ambiguity -- words may have several meanings, and the correct one cannot be chosen based solely on the word itself.
  - bank in English
- Solution -- resolve the ambiguity on the spot by POS tagging (if possible) or pass-on the ambiguity to the other levels.

Syntactic Processing

- Syntactic analysis involves the analysis of words in a sentence for grammar and arranging words in a manner that shows the relationship among the words.
- The purpose of this processing is two folds: to check that a sentence is well formed or not and to break it up into a structure that shows the syntactic relationships between the different words
- **Parsing --** converting a flat input sentence into a hierarchical structure that corresponds to the units of meaning in the sentence.
- There are different parsing formalisms and algorithms.
- Most formalisms have two main components:
  - **grammar --** a declarative representation describing the syntactic structure of sentences in the language.
  - **parser --** an algorithm that analyzes the input and outputs its structural representation (its parse) consistent with the grammar specification.
- CFGs are in the center of many of the parsing mechanisms. But they are complemented by some additional features that make the formalism more suitable to handle natural languages.

Semantic Analysis

- Assigning meanings to the structures created by syntactic analysis.
- Mapping words and structures to particular domain objects in way consistent with our knowledge of the world.
- Semantic can play an import role in selecting among competing syntactic analyses and discarding illogical analyses.
  - `I robbed the bank` -- bank is a river bank or a financial institution
- We have to decide the formalisms which will be used in the meaning representation.

Knowledge Representation for NLP

- Which knowledge representation will be used depends on the application -- Machine Translation, Database Query System.
- Requires the choice of representational framework, as well as the specific meaning vocabulary (what are concepts and relationship between these concepts -- ontology)
- Must be computationally effective.
- Common representational formalisms:
  - first order predicate logic
  - conceptual dependency graphs
  - semantic networks
  - Frame-based representations

Discourse

- Disclosure integration takes into account the context of the text.
- Discourses are collection of coherent sentences (not arbitrary set of sentences)
- Discourses have also hierarchical structures (similar to sentences)
- **anaphora resolution --** to resolve referring expression
  - `Mary bought a book for Kelly.` **`She`** `didn’t like` **`it`**`.`
    - **She** refers to Mary or Kelly. -- possibly Kelly
    - **It** refers to what -- book.
  - `Mary had to lie for Kelly.` **`She`** `didn’t like` **`it`**`.`
- Discourse structure may depend on application.
  - Monologue
  - Dialogue
  - Human-Computer Interaction

Natural Language Generation

- NLG is the process of constructing natural language outputs from non-linguistic inputs.
- NLG can be viewed as the reverse process of NL understanding.
- A NLG system may have two main parts:
  - **Discourse Planner --** what will be generated. which sentences.
  - **Surface Realizer --** realizes a sentence from its internal representation.
- **Lexical Selection --** selecting the correct words describing the concepts.

Machine Translation

- Machine Translation -- converting a text in language A into the corresponding text in language B (or speech).
- Different Machine Translation architectures:
  - interlingua based systems
  - transfer based systems
- How to acquire the required knowledge resources such as mapping rules and bi-lingual dictionary? By hand or acquire them automatically from corpora.
- Example Based Machine Translation acquires the required knowledge (some of it or all of it) from corpora.

NLP Applications

- Machine Translation – Translation between two natural languages.
- Information Retrieval – Web search (uni-lingual or multi-lingual).
- Query Answering/Dialogue – Natural language interface with a database system, or a dialogue system.
- Report Generation – Generation of reports such as weather reports.
- Some Small Applications – Grammar Checking, Spell Checking, Spell Corrector
- Sentiment Analysis
- Language Detection
- Speech Recognition
- Automatic Text Summarization
- Named Entity Recognition
- Text Generation
- Part-of-speech (POS) Tagging
- ChatBot

Advantages of NLP

- NLP system offers exact answers to the questions, no unnecessary or unwanted some information.
- The accuracy of the answer increases with the amount of relevant information provided in the questions.
- Structuring a high unstructured data source.
- Users can ask questions about any subject and get a direct response in seconds.
- It is easy to implement.
- Using a program is less costly than hiring a person. A person can take two or three times longer than a machine to execute the tasks mentioned.
- NLP system provides answers to the questions in natural language.

- Allow you to perform more language-based data compares to a human being without fatigue and in an unbiased and consistent way.
- NLP process help computer communicate with a human in their language and scales other language-related tasks.
- It is a faster customer service response time.

Disadvantages of NLP

- NLP system doesn’t have a user interface that lacks features that allow users to further interact with the system.
- If it is necessary to develop a model with a new one without using a pre-trained model, it can take a week to achieve a good performance depending o the amount of data.
- The system is built for a single and specific task only, it is unable to adapt to new domains and problems because of limited functions.
- In complex query language, the system may not be able to provide the correct answer it a question that is poorly worded or ambiguous.
- It is not 100% reliable, It is never 100% dependable. There is the possibility of error in its prediction and results.
- NLP is unpredictable
- NLP is unable to adapt to the new domain, and it has a limited function that's why NLP is built for a single and specific task only.